

João Vitor Bermal Santaniello

Software Engineer · Full-Stack, AI-Augmented Delivery and Geospatial Data

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SUMMARY

Software engineer with six years of production experience across water utilities, energy, healthcare and telecom. I build full-stack products and data pipelines with FastAPI, Node.js, React, Next.js, Python and ArcGIS. I was the main developer of EyeConnect, a teleophthalmology platform in production, from its clinic DICOM gateway to OCR exam intake, rewrote ISPDrive, a multi-tenant SaaS for internet providers, and build the geospatial analyses behind expansion studies at Aegea, one of the largest sanitation groups in Brazil. Claude Code is now my primary environment. MBA in Software Engineering (USP/ESALQ, deep learning thesis graded 9/10), fluent English with US teams.

EXPERIENCE

Software Developer, Bizpoke Soluções em Software · São José dos Campos, SP

Dec 2022 to Present

Aegee · Expansion planning and Regenera

- Cut a recurring geospatial workflow from more than 3 hours to about 20 minutes with Spatial Extractor, a reusable Python and GeoPandas tool with parallel spatial joins, validation and Excel/Shapefile export.
- Delivered census tract indicators for 16 Brazilian states by crossing IBGE census, CNEFE addresses and ANEEL BDGD records, giving the expansion team demand estimates for concession studies.
- Co-designed the CITIUS/Regenera data model for landfill biogas operations and loaded its historical data (wells, drains, collectors, readings), feeding dashboards with measurement beacons and normalized CH4 classes.
- Kept production dashboards running through the ArcGIS Enterprise 11.5 migration by auditing and fixing panels, and wrote a governance script that reports unused portal items for cleanup.

AI-augmented delivery with Claude Code

- Authored a shared library of nine Claude Code skills and a root CLAUDE.md inherited by 13 Aegee project folders (project bootstrap, ArcGIS API client, dashboard editing, delivery documents, status reporting), so every teammate works from the same playbook.
- Connected Claude Code through MCP to ClickUp for daily client status reports, and contributed to BizpokeBI, our geospatial data platform, and its bizpoke MCP server with approval-gated writes.

EyeConnect · teleophthalmology platform in production

- Built, as sole developer, a Raspberry Pi exam gateway for clinics: Orthanc receives DICOM from eye devices, and a Node.js service classifies exams, reads PDF reports with AWS Textract and posts them to the platform over OpenVPN.
- Brought the OCR into the platform: bulk PDF uploads to S3, AWS Textract and parsers for 14 exam devices (Zeiss Cirrus and HFA, Spectralis and others), with deduplication and retries.
- Generated rule-based draft reports for visual field and OCT exams and structured reports with PDF output, and integrated the Clinic Web (Linx) API to find patients by CPF and send exams back.

ISPDrive · white-label cloud storage for internet providers

- Rewrote the product in FastAPI, MongoDB and Next.js 14 as a multi-tenant SaaS: per-provider branding, sign-in against each provider's ERP, and uploads and downloads straight to DigitalOcean Spaces through presigned URLs.
- Built the admin area with usage dashboards, an audit trail of 11 action types and a quota report used for invoicing, and deployed it on DigitalOcean with Docker Compose, NGINX and automatic Let's Encrypt renewal.

Equatorial Energia · DOMO field app

- Led front-end and mobile delivery for DOMO, building an offline-capable map component in React Native (Expo) with feature create, edit and delete, attribute forms, extra layers and tablet support.
- Prototyped the project validation flow in Figma and implemented it in Next.js, TypeScript, Redux and MUI after sign-off from Equatorial.

US utilities through SBS

- Automated GIS workflows for North Las Vegas and Duke Energy and integrated SAP, IBM Maximo and Schneider Electric via Python and JS SDKs.

AI Coding Benchmark Contributor (freelance), Alignerr · Remote

May 2025 to Present

- Wrote Python problems with reference solutions and test suites for LiveCodeBench and BigCodeBench, each designed to break at least 2 of 4 frontier models, and reviewed prompts and grading criteria in the Mango and Fairylights programs.

Software Development Intern, Bizpoke Soluções em Software · São José dos Campos, SP

Apr 2021 to Dec 2022

- Shipped React, Node.js, Python and C# features for client projects and produced spatial data with ArcGIS Pro and Enterprise alongside offshore teams, in English.

Web Development Intern, NFe Sistemas · São José dos Campos, SP

Oct 2020 to Mar 2021

- Delivered front-end and back-end changes with HTML, CSS, JavaScript, jQuery, PHP and MySQL.

SELECTED PROJECTS

Triage Desk, AI support triage for D2C brands · 2026

trriage.joaosantaniello.com

Live demo of AI customer support for several brands: n8n ingests messages, a Claude agent in a Supabase Edge Function drafts replies with policy and catalog sub-agents, and staff approve them in Next.js. Brand data is isolated with Postgres RLS and pgTAP tests.

Code: github.com/joao-bermal/triage-desk

Retinal AVR pipeline, MBA thesis in deep learning · 2025 to 2026

GitHub

PyTorch pipeline that estimates the arteriolar-to-venular ratio, a cardiovascular risk marker, from retinal photographs: a U-Net segments vessels (Dice 0.79 on held-out DRIVE images), a ResNet-50 network separates arteries from veins, and an optic disc model cut the disc center error from 358 to 11 px for Knudtson caliber measurement.

Miau Atelier, Founder, designer and developer · 2026

miauatelier.com

A cat furniture brand for the US market, built end to end: brand identity, Shopify store in USD with Stripe, Python automation over the Admin GraphQL API and an AI image pipeline for product photography.

NAMMAN, Next.js on Vercel · 2026

namman.vercel.app

Browser-only app that syncs Neural Amp Modeler profiles from the TONE3000 API to disk: OAuth 2.0 with PKCE, File System Access API, IndexedDB, backoff and throttled bulk downloads, no server.

EDUCATION

MBA in Software Engineering, Universidade de São Paulo (USP/ESALQ)

2024 to 2026

- Coursework in back-end and front-end architecture, APIs and micro-frontends, cloud and Kubernetes, clean architecture andDDD, machine learning and model deployment, observability and TDD.
- Thesis graded 9/10: the retinal AVR pipeline under Selected projects.

Associate degree in Systems Analysis and Development, Universidade Paulista (UNIP)

2021 to 2022

- Software design, databases, web and back-end development, completed while working full time.

Technical Program in Computing, with high school, Colégio Técnico Antônio Teixeira Fernandes

2018 to 2020

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Bash

Data and GIS: GeoPandas, ArcGIS Pro, ArcGIS Enterprise 11.5, ArcGIS Dashboards, Experience Builder, ArcGIS JS API, FME, PostgreSQL, enterprise geodatabases, ETL and data modeling

Back end: FastAPI, Node.js, Meteor, REST APIs, JWT, OAuth, WebSockets, MongoDB, PostgreSQL, Supabase, DICOM

Front end and mobile: React, Next.js, Redux Toolkit, Tailwind CSS, MUI, React Native (Expo), Figma

Cloud and DevOps: Docker, NGINX, Linux, AWS (S3, Textract, EC2), DigitalOcean, Vercel, n8n, basic Kubernetes

AI and ML: Claude Code (skills, MCP, sub-agents), Claude API, PyTorch, OpenCV, OCR (Textextract), LLM evaluation

E-commerce and design: Shopify (Admin GraphQL, Liquid), Stripe, brand identity, generative image pipelines

Ways of working: Scrum, Git and GitFlow, Jira, ClickUp, technical documentation, remote work with US teams

LANGUAGES · COURSES AND EVENTS

Portuguese (native) · English (fluent, daily professional use)

Python 3: Deep Dive, Part 1 (Udemy, 2025) · Deep Learning Using ArcGIS (Esri, 2024) · AWS Summit São Paulo (2023)